

Cyber Tabletop Exercise

ATM Switch Outage and Cash Access Disruption

Scenario source: Structured from the uploaded STIX bundle describing a retail bank operating a national ATM fleet, the targeted ATM switch and cash access services, the incident "ATM network disruption during holiday weekend," and two scenario notes covering national failed withdrawals and public trust/liquidity decisions.

1. Exercise Overview

Exercise title: ATM Switch Outage and Cash Access Disruption

Exercise type: Facilitated tabletop exercise (TTX)

Audience: Executive, operational, technical, communications, customer operations, fraud, branch operations, treasury/liquidity, legal/compliance

Recommended duration: 90 minutes

Scenario classification: Internal use only

Purpose

This tabletop exercise prepares participants to respond to a disruptive cyber incident affecting ATM transaction routing and customer cash access during a high-demand holiday weekend. The scenario is designed to test coordination across crisis management, technology operations, customer support, communications, and regulatory response.

Scenario basis

The underlying STIX scenario describes:

- a **banking organization** in the **financial-services** sector operating a **national ATM fleet**;
 - critical **ATM switch and cash access services** supporting routing, authorization, and availability monitoring;
 - an **attack pattern** focused on disrupting ATM transaction routing and cash access;
 - an **incident** involving an ATM network disruption during a holiday weekend;
 - initial injects involving failed withdrawals, duplicate debit holds, branch overflow, public rumors, emergency cash support, fee remediation, and communications decisions.
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2. Exercise Objectives

By the end of the exercise, participants should be able to:

1. Assess and coordinate the initial response to a suspected cyber-related outage affecting ATM availability at national scale.
 2. Prioritize customer impact management, including cash access alternatives, fee remediation, branch support, and call-center surge handling.
 3. Evaluate crisis governance, escalation thresholds, and executive decision-making during a holiday-period disruption.
 4. Test internal and external communications, including customer notifications, media posture, processor coordination, and regulator engagement.
 5. Validate technical containment and recovery decision-making for ATM switch services, including restoration prerequisites and monitoring.
 6. Identify gaps in business continuity, liquidity management, operational resilience, and cyber incident playbooks.
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3. Scope and Assumptions

In scope

- ATM withdrawals, balance inquiries, and transaction routing
- ATM switch authorization and availability monitoring
- Branch customer overflow and exception handling
- Contact center escalation and customer complaint management
- Treasury and liquidity considerations for emergency cash support
- Communications with customers, media, regulators, payment partners, and processors
- Technical containment, investigation, and recovery governance

Out of scope

- Detailed malware reverse engineering
- Full legal determination of breach notification obligations by jurisdiction
- Vendor contract negotiation specifics

Assumptions

- The organization is a retail bank with nationwide ATM dependence.
 - The incident occurs during a holiday weekend with elevated cash demand and reduced staffing.
 - No confirmed destructive malware is identified at exercise start.
 - Initial information is incomplete and may evolve.
 - Participants should answer according to existing roles, responsibilities, and real operating procedures.
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4. Participant Roles

Executive / Crisis leadership

- Executive sponsor
- Crisis management lead
- Business line leadership
- Treasury / liquidity representative
- Legal / compliance lead

Operational / business

- Branch operations
- Contact center / customer operations
- ATM operations / vendor management
- Payments operations
- Fraud operations
- Communications / media relations

Technical / cyber

- SOC / cyber incident commander
 - Infrastructure / network operations
 - Application owner for ATM switch services
 - Third-party risk / vendor management
 - Resilience / disaster recovery lead
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5. Rules of Engagement

- This is a no-fault learning exercise.
 - There are no perfect answers; the goal is to surface assumptions, decision points, and improvement opportunities.
 - Treat the scenario as real and respond using current organizational processes.
 - Real incidents take precedence over the exercise.
 - Facilitators may introduce new information to test adaptability.
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6. Scenario Narrative

At the start of a long holiday weekend, customers across multiple regions begin reporting failed ATM withdrawals. Some customers also report duplicate debit holds after unsuccessful transactions. Social media complaints rise rapidly, and branches see increased walk-in traffic as customers seek access to cash and account confirmation.

The bank's ATM switch and cash access services are central to transaction routing, authorization, and service availability monitoring. The disruption is initially treated as a service outage, but cyber involvement is suspected as indicators point to a targeted effort to disrupt ATM transaction routing and degrade customer access to cash.

As the incident develops, the bank faces several parallel pressures: restoring essential services, preserving customer trust, managing branch and call-center surge, coordinating with processing partners, and making timely decisions on emergency access to cash, fee refunds, and public communications.

7. What Is the ATM Switch and Cash Access Service?

The ATM switch and cash access environment is the service layer responsible for:

- routing ATM transactions to the correct authorization path;
- validating and processing withdrawal and balance requests;
- supporting availability monitoring and incident visibility;
- coordinating with processors and dependent banking services;
- enabling nationwide cash access through owned and partner ATM channels.

A disruption in this environment can create widespread customer-facing impact even when core banking remains partially available.

8. Exercise Timeline and Injects

Inject 1 – National spike in failed withdrawals

Simulated time: Friday, 08:00

Elapsed time: Incident discovery

Inject

Customers across multiple regions report failed ATM withdrawals and duplicate debit holds, while branch call volumes surge. Internal monitoring shows a sharp increase in ATM authorization failures and repeated timeout conditions in the ATM switch service.

What participants know

- ATM failures are occurring nationally, not in a single region.
- Some transactions appear unsuccessful to customers but still create temporary debit holds.
- Branches are beginning to receive overflow traffic.
- The root cause is not yet confirmed.

Discussion questions

- How is the incident initially classified and who owns command?

- What threshold triggers crisis management activation?
- What immediate actions are taken to verify scope, customer impact, and dependency failures?
- What customer-facing channels need rapid updates first?
- How will the bank handle duplicate holds and potential fee remediation from the outset?
- What surge actions are needed for branches and contact centers?

Expected actions

- Open and triage a high-severity incident.
- Confirm business impact across channels and regions.
- Activate incident command and notify key executives.
- Begin customer service guidance for known symptoms.
- Prepare branch exception procedures and customer support scripts.
- Establish a decision log and common operating picture.

Facilitator notes

Push participants to distinguish between a technology outage and a possible cyber incident. Challenge assumptions around whether failed withdrawals, duplicate holds, and branch overflow are handled by existing business continuity plans.

Inject 2 – Public trust, cash rationing, and liquidity decisions

Simulated time: Friday, 10:30

Elapsed time: 2.5 hours after discovery

Inject

The bank must decide how to ration emergency cash support, adjust fee refunds, publish status updates, and coordinate with the processor while rumors spread online. Media inquiries begin asking whether the outage is cyber-related and whether customer balances are at risk.

What participants know

- Public complaints are accelerating online.
- Rumors suggest either a cyberattack or a broader banking systems failure.
- Call-center wait times are rising sharply.
- Some branches report elevated demand for over-the-counter withdrawals.
- Processor coordination is ongoing, but restoration timing remains uncertain.

Discussion questions

- Who decides whether to enable emergency cash support and under what limits?
- How should the bank prioritize access to cash for vulnerable or high-need customer groups?
- What is the approach to refunding fees, duplicate holds, and customer remediation?
- What should the first public statement say, and what should it avoid saying?

- When do regulators, payment partners, law enforcement, or sector coordination groups need notification?
- How do treasury, operations, and branch leadership coordinate if demand for cash rises materially?

Expected actions

- Stand up cross-functional crisis management.
- Define emergency cash-access thresholds and exception authority.
- Approve customer remediation principles.
- Issue a clear, plain-language customer status update.
- Coordinate with processor/vendor and document external dependencies.
- Prepare regulator notification decision points.

Facilitator notes

Probe whether participants can balance operational prudence with customer trust. Push for concrete decisions: fee waivers, messaging cadence, branch staffing, partner ATM arrangements, and escalation to executive leadership.

Inject 3 – Cyber suspicion intensifies and service restoration is delayed

Simulated time: Friday, 14:00

Elapsed time: 6 hours after discovery

Inject

Cyber operations identify anomalies consistent with targeted disruption of transaction routing logic or supporting service dependencies. There is still no evidence of customer account compromise, but restoration remains uncertain. Branches report rising confrontational customer interactions, and the contact center is exceeding service-level targets.

What participants know

- The event is now treated as a suspected cyber-related service disruption.
- No confirmed data loss exists at this stage.
- Restoration may require staged recovery and heightened monitoring.
- Customer frustration is translating into reputational and safety concerns at some sites.

Discussion questions

- What technical containment decisions are appropriate before recovery is attempted?
- What evidence is needed before reconnecting or restoring ATM switch functions?
- How should the bank message the difference between service disruption and account compromise?
- What safety, fraud, and conduct risks are increasing as the outage continues?
- What workarounds remain viable if the outage extends through the weekend?

Expected actions

- Move to cyber incident governance.
- Preserve evidence and validate the recovery path.
- Implement enhanced fraud monitoring and branch security coordination.
- Update customer messaging and internal talking points.
- Prioritize essential services and define minimum viable restoration.

Facilitator notes

Focus discussion on risk tradeoffs: restoring quickly versus restoring safely. Encourage technical and business participants to articulate recovery prerequisites in business language.

Inject 4 – Service restoration, backlog management, and post-incident obligations

Simulated time: Saturday, 09:00

Elapsed time: 25 hours after discovery

Inject

The processor confirms stability has been restored, but the bank's cyber and technology teams require validation checks before full reconnection and normalization. Transaction backlogs, customer complaints, fee reversals, and branch cash replenishment issues remain active. Senior leadership requests a preliminary impact summary and regulator-ready timeline.

What participants know

- Core disruption conditions are improving.
- Reconnection and restoration must be controlled and monitored.
- Customer remediation and complaint handling will continue after technical restoration.
- A post-incident report will likely be required.

Discussion questions

- Who authorizes reconnection and on what criteria?
- How will the bank validate transaction integrity after restoration?
- What is the plan for backlog processing and customer remediation?
- What lessons learned should be captured immediately while facts are fresh?
- What communications are needed to close the incident with customers, regulators, and internal stakeholders?

Expected actions

- Execute staged restoration with clear approval gates.
- Validate transaction integrity, debit hold reversals, and balance correctness.
- Publish recovery communications and remediation steps.
- Prepare executive summary, regulator narrative, and after-action review.

- Assign owners for lessons learned and resilience improvements.

Facilitator notes

Ensure participants address both technical recovery and customer recovery. The incident is not over when the service returns; trust, remediation, and regulatory follow-through matter equally.

9. Decision Points

Facilitators should ensure the group reaches explicit decisions on:

- incident severity and escalation level;
 - crisis management activation;
 - customer communications timing and approval authority;
 - fee waivers, duplicate hold remediation, and hardship support;
 - emergency cash access channels and rationing criteria;
 - regulator notification thresholds;
 - recovery approval gates and reconnection authority;
 - post-incident review ownership and timeline.
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10. Business and Technical Impacts

Business impacts

- Customer inability to access cash during a peak-demand period
- Branch congestion and staff strain
- Contact-center service degradation
- Reputational damage driven by visible consumer impact
- Potential regulatory scrutiny over operational resilience and customer treatment
- Increased treasury and cash logistics pressure

Technical impacts

- ATM switch transaction-routing instability
 - Authorization failures and timeout conditions
 - Monitoring and dependency validation challenges
 - Need for staged recovery, enhanced logging, and integrity checks
 - Elevated fraud monitoring requirements during service degradation
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11. Expected Good Practices

Participants should demonstrate:

- rapid formation of a common operating picture;
 - clear crisis ownership and decision logging;
 - customer-centric remediation planning;
 - close coordination across cyber, IT, operations, treasury, and communications;
 - evidence-based recovery governance;
 - early preparation for regulator and board-level reporting.
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12. Debrief Questions

- What worked well in the first 60 minutes of the scenario?
 - Where did roles, responsibilities, or escalation thresholds appear unclear?
 - Did the bank have a practical emergency cash-access model for a national outage?
 - Were customer communications timely, accurate, and empathetic?
 - Did restoration governance appropriately balance speed and safety?
 - What capabilities, playbooks, or partner arrangements should be improved before the next exercise?
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13. After-Action Improvement Areas

Potential improvement themes include:

- ATM outage and cash-access playbooks
 - processor and third-party coordination procedures
 - duplicate hold and fee-remediation workflows
 - branch surge and security planning
 - regulator notification matrices
 - crisis communications templates for retail-service disruptions
 - recovery validation criteria for transaction integrity
 - operational resilience testing for holiday and peak-demand periods
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14. Facilitator Quick Reference

Suggested pace

- Introduction and objectives: 10 minutes
- Inject 1: 15 minutes
- Inject 2: 20 minutes
- Inject 3: 20 minutes
- Inject 4: 15 minutes

- Debrief and lessons learned: 10 minutes

Artifacts facilitators may optionally use

- Mock customer complaints and social posts
 - Branch queue and call-center volume dashboards
 - Processor status email update
 - Draft media statement for review
 - Regulator inquiry email
 - Executive situation report template
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15. MITRE ATT&CK Mapping

The uploaded STIX bundle explicitly identifies an attack pattern focused on disrupting ATM transaction routing and cash access. Because the STIX does not provide a detailed intrusion chain, the ATT&CK mapping below reflects the **scenario behaviors implied by the disruption objective**, not confirmed forensic findings.

Primary ATT&CK tactics reflected

- **Impact (TA0040)** – The core scenario objective is service disruption affecting customer cash access and ATM transaction availability.
- **Command and Control / Persistence / Lateral Movement** – Not evidenced in the STIX and therefore not asserted as confirmed.

Most relevant ATT&CK techniques reflected by scenario outcome

- **T1499 – Endpoint Denial of Service** or **T1498 – Network Denial of Service**: plausible if the disruption is caused by overwhelming or destabilizing service components.
- **T1489 – Service Stop**: plausible if ATM switch services or supporting transaction-routing services are intentionally disabled.
- **T1496 – Resource Hijacking**: not indicated.
- **T1565 – Data Manipulation**: plausible only if routing logic, transaction state, or authorization responses were altered to create duplicate holds or failed transactions; not confirmed.

Analyst note

The strongest defensible ATT&CK alignment from the provided STIX is **Impact**, especially service disruption against a customer-facing financial service. More granular technique mapping should be refined if additional intrusion details are later provided, such as access method, tooling, or confirmed service manipulation behavior.

16. Source Traceability

This tabletop exercise was generated from the uploaded STIX bundle and informed by the structure used in the sample tabletop exercise documents you provided.